Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
2	*(a)*		Angle when arc [length] equals radius Accept about 57.3° **or** angle when 2π = 360° or cycle / circle	1			1		
	(b)	(i)	Use of $T = \frac{1}{f}$ (1) Answer = 1.67 [s] (1) Accept $\frac{5}{3}$ **or** 1.66 **or** $1.\dot{6}$ **or** 1.7 [s] Don't accept 1.6 [s]	1	1		2	1	
		(ii)	Substitution into $\omega = \frac{2\pi}{T}$ or $2\pi f$ or **and** $v = \omega r$ (1) **ecf** on T or f v = 10.6 [m s^{-1}] (1) (Accept 10.5 m s^{-1} if 1.67 s used)		2		2	2	
	(c)		$N = \frac{mv^2}{r}$ or $mr\omega^2$ or implied (1) $N = \frac{66.2 \times (10.6)^2}{2.8}$ = [2 634] [N] (1) **ecf** on v and ω accept approximately 2 657 [N] F = 66.2 × 9.81 = [649.4 [N]] (1) Vertical forces are balanced or equivalent e.g. $F = W$ (1)	1 1	1 1		4	2	
	(d)	(i)	650 ≤ **or** = **or** < μ × 2 600 (1) So $\mu >$ **or** = 0.24 or 0.25 (1) **Alternative:** 2 600 × 0.25 (1) = 650 (1)	1	1		2	2	
		(ii)	Friction = 650 [N] or implied (1) $\frac{650}{0.45}$ = 1 444 [N] (1) Equating to centripetal (1) ω = 2.8 [rad s^{-1}] (1) Answer of 2.51 [rad s^{-1}] award 1 mark only		1	1 1 1	4	2	
			Question 2 total	**5**	**7**	**3**	**15**	**9**	**0**